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VII) Vaccum & Storage Management

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Section 7 : Vaccum & Storage Management

A) What VACCUM Actualy does :

problem :

- Delta lake never deletes files immediately
- overtime tables accumulate many obsolete f.

VACCUM solves this problem.

→ removes obsolete files that are no longer needed

VACCUM only deletes files that are :
- Not referenced by any snapshot
- And older than retention period

B) Retention Period :

(minimum time Delta keeps old data files before they are eligible for deletion)

↳ By default : 7 days (168 hours)

↳ so when we run :

```
VACCUM sales;
```

↳ Delta actually run;

```
VACCUM sales RETAIN 168 HOURS
```

Overriding Retention :

```
SET spark.databricks.delta.retentionDurationCheck
    .enabled = false;
VACCUM sales RETAIN 1 HOURS.
```

c) Why VACCUM can be Dangerous?

→ Because; it permanently deletes files from the storage.

→ No recovery.

→ If we get try to access deleted versions we get : (FileNotFoundException)

Restore of that version will be Impossible.

D) Storage cleanup process:

- Scan the txn log
- Identify Active files
- Identify obsolete files
- Apply retention rule
- Delete safe files

Log cleanup

We have for every 10 commits; we have a Checkpoint

↓

So;

once checkpoints exists, Delta can safely remove very old JSON logs

VACCUM (vs) Log cleanup:

VACCUM	old datafiles will be removed
Log cleanup	old txn logs will be removed